Nikkila Prakash

Edinburgh, UK | LinkedIn

Education

University of Edinburgh

Master of Science in Computer Science *Sep. 2026 – Aug. 2027*

- Relevant modules: Machine Learning, Distributed Systems, AI & ML Systems Deployment

Experience

Software Engineer Jul. 2023 – Jul. 2026

PayPal | Python, LLM Fine-tuning, RAG, MCP, Java, Spring Boot, Kafka, BigQuery

- Built an LLM-powered conversational interface using GPT-4 and the Model Context Protocol for privacy queries, enhancing user ticket resolution times by 15% and enabling self-service workflows.
- Developed an automated compliance validator with LLM-based policy interpretation which reduced legal processing time by 60%.
- Designed and developed a Spring Boot decisioning system processing 10M daily requests, implementing distributed caching to achieve a 20% latency reduction across global operations.
- Designed and implemented core components for a low-latency Consent Management platform serving PayPal, Venmo, and Honey across the Americas, Europe, and APAC for 400M+ users, enforcing consent with real-time policy evaluation across multiple jurisdictions.
- Led 2 junior engineers as technical SPOC for 17 consent types, architecting reusable frameworks that reduced requirement onboarding time by 40% across the product ecosystem.
- Developed a REST API validation layer handling 500K daily calls for account creation, upgrades, and merchant data utilization, ensuring compliance across the PayPal extended ecosystem.

Software Engineering Intern Aug. 2022 – Jan. 2023

PayPal | Scala, Apache Spark, Web Crawling

- Built a web crawler automation script using Apache Spark and Scala, which enhanced system efficiency by 30%, processed 500K+ URLs for large-scale cookie classification supporting European GDPR compliance.

Patents & Publications

Autonomous Multi-Agent System for Production Errors *Patent Filed 2025*

- Primary claims cover an LLM-powered multi-agent workflow for autonomous error detection, classification, and resolution.

Cardiovascular Disease Risk Assessment Using Machine Learning *IEEE Xplore ICICT'23, 2023*

- First-author work applying supervised learning algorithms to cardiovascular risk prediction. [IEEE Xplore]

Technical Skills

AI: LLM Fine-tuning, Multi-Agent Architectures, Retrieval Augmented Generation, Model Context Protocol
Programming Languages: Java, Python, C++, Scala, JavaScript, SQL, HTML/CSS
Backend Frameworks & APIs: Spring Boot, JDBC, REST APIs, GraphQL
Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda, SageMaker, Aurora), Docker, Load Balancing, Auto-Scaling
Databases & Storage: MySQL, MongoDB, Oracle DB, BigQuery, JunoDB
DevOps & Tools: Git/GitHub, Jenkins, CI/CD Pipelines, SonarQube, JUnit, Test-Driven Development, Agile/Scrum
Distributed Systems & Big Data: Kafka, Microservices, Apache Spark, Event-Driven Architecture, Distributed Caching, Message Queuing (AMQ)

Achievements

- Code.AI Hackathon, Winner (1st) — multi-agent privacy compliance solution combining LLM-based policy interpretation with automated compliance validation.
- PayPal Recognition — received 12 internal awards for innovation, technical excellence, and collaboration.
- Innovation Chapter, Core Member — drove strategic technology exploration and built prototypes for various patent initiatives.
- Technical Mentor — guided 5 junior engineers (3 interns, 2 new graduates) through onboarding, code reviews, and career development.